

Qinglin Zhu

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PROFILE

PhD candidate in NLP at King's College London. My research focuses on improving the generation, reasoning, and code intelligence of large language models, with recent work on *repository-level code editing*, *latent-space reasoning*, and *retrieval-augmented generation*. I have hands-on experience in large-scale data construction and LLM training on hundreds of GPUs, with industry experience at MSRA, Baidu, Shopee, and Tencent Music.

EDUCATION

- **King's College London** London, UK
Ph.D. in Computer Science (NLP). Supervisors: Prof. Lin Gui, Prof. Yulan He. Oct. 2023 - Present
- **Harbin Institute of Technology** Shenzhen, China
M.Sc. - Computer Science and Technology; GPA: 3.667/4.0; Rank: 1/69 Sept. 2020 - Mar. 2023
- **Harbin Institute of Technology** Shenzhen, China
B.Eng. - Computer Science and Technology; GPA: 3.773/4.0; Rank: 14/70 Sept. 2016 - Jun. 2020

SELECTED PUBLICATIONS

* denotes equal contribution

LLM Reasoning & Code Intelligence

- **Qinglin Zhu**, Tianyu Chen, Shuai Lu, Lei Ji, Runcong Zhao, et al. *Pull Requests as a Training Signal for Repo-Level Code Editing*. **ICML'2026**.
- **Qinglin Zhu**, Runcong Zhao, Hanqi Yan, et al. *Soft Reasoning: Navigating Solution Spaces in Large Language Models through Controlled Embedding Exploration*. **ICML'2025 (Spotlight)**.
- Hanqi Yan*, **Qinglin Zhu***, Xinyu Wang, et al. *Mirror: A Multiple-perspective Self-Reflection Method for Knowledge-rich Reasoning*. **ACL'2024**.
- **Qinglin Zhu**, Yizhen Yao, Runcong Zhao, et al. *Latent Refinement Decoding: Enhancing Diffusion-Based Language Models by Refining Belief States*. Submitted to **NeurIPS'2026**.

Knowledge-Rich Reasoning, Retrieval & Memory

- Runcong Zhao*, **Qinglin Zhu***, Hainiu Xu, et al. *SymbolicThought: Integrating Language Models and Symbolic Reasoning for Consistent and Interpretable Human Relationship Understanding*. **ACL'2026 Demo**.
- Runcong Zhao*, **Qinglin Zhu***, Hainiu Xu, et al. *Large Language Models Fall Short: Understanding Complex Relationships in Detective Narratives*. **ACL'2024 Findings**.
- Zhanghao Hu, **Qinglin Zhu**, Siya Qi, et al. *Spectrum Projection Score: Aligning Retrieved Summaries with Reader Models in RAG*. **AAAI'2026 (Oral)**.
- Zhanghao Hu*, **Qinglin Zhu***, Runcong Zhao, et al. *Beyond RAG for Agent Memory: Retrieval by Decoupling and Aggregation*. Submitted to **NeurIPS'2026**.

Previous work on Stance Detection & Sentiment Analysis:

- [SIGIR'2022] **Qinglin Zhu**, et al.; [ACL'2022] Bin Liang*, **Qinglin Zhu***, et al.; [SemEval'2021] **Qinglin Zhu**, et al.; [CCL'2020] **Qinglin Zhu**, et al.

RESEARCH & INDUSTRY EXPERIENCE

- **Microsoft Research Asia (MSRA)** Beijing, China
Research Intern May. 2025 - Present
 - **Repo-Level Code Editing** (May 2025 - Feb. 2026; **ICML'26**):
 - * Proposed **Clean-PR**, a scalable mid-training paradigm that turns noisy GitHub pull requests into verified Search/Replace edit blocks through filtering, reconstruction, and round-trip validation; built the largest publicly available corpus of **2M PRs** across 12 programming languages.
 - * Conducted LLM mid-training on 100+ GPUs and Agentless-aligned SFT with error-driven data augmentation, achieving **+13.6%** on SWE-bench Lite and **+12.3%** on SWE-bench Verified.
 - * Demonstrated that repository-level editing capability can be internalized into model weights and transferred to Multi-SWE-bench Flash (7 languages) and Agent based method(OpenHands).
- **King's College London** London, UK
PhD Researcher Oct. 2023 - Present
 - **Latent-Space Reasoning and Decoding**:
 - * Proposed **Soft Reasoning (ICML'25 Spotlight)**, a training-free inference-time framework that explores solution spaces by optimizing continuous perturbations of the first answer-token embedding through low-dimensional Bayesian optimization.

- * Extended this latent-space perspective to diffusion language models through **Latent Refinement Decoding**, preserving uncertain predictions as soft belief states and progressively refining them via predictive feedback and convergence-aware token commitment.
- **Knowledge-Rich, Symbolic, and Agentic Reasoning:**
 - * Developed a line of structured reasoning methods spanning **Mirror** (ACL'24), **Conan** (ACL'24 Findings), **SymbolicThought** (ACL'26 Demo), and **PLAYER***, progressing from multi-perspective self-reflection and complex relational reasoning to symbolic constraint enforcement and agentic planning in interactive narratives across increasingly complex knowledge-rich reasoning settings.
- **Retrieval and Agent Memory:**
 - * Co-developed retrieval and memory methods across increasingly complex information-access settings, spanning **EmbQA** (ACL'25), **SPS/xCompress** (AAAI'26 Oral), and **xMemory**, progressing from embedding-level query refinement and reader-guided retrieval compression to hierarchical memory decomposition and top-down aggregation for accurate and token-efficient knowledge access.

Baidu

Shanghai, China

Advertising Algorithm Engineer (Full-time)

Apr. 2023 – Oct. 2023

- Developed and optimized the Phoenix Nest advertising system's central dispatch unit ("Mixer") for online ad allocation, focusing on low-latency ranking/dispatch logic, experiment iteration, and user-preference-aware e-commerce title refinement.

Shopee

Shanghai, China

NLP Algorithm Engineer (Intern)

Jul. 2022 – Sept. 2022

- Built sentiment knowledge-enhanced pre-training and active learning pipelines for e-commerce review understanding, improving aspect sentiment triplet extraction by +8.6% F1 and reducing annotation pressure through targeted sample selection.

Tencent Music Entertainment Group

Shenzhen, China

NLP Algorithm Engineer (Intern)

Aug. 2021 – Apr. 2022

- Led aspect-based sentiment analysis for online music content, designing hierarchical classification and active learning workflows to improve training data quality and model iteration efficiency.

HONORS AND AWARDS

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|---|-----------|
| • Outstanding Master's Graduate & Outstanding Master's Thesis, <i>HIT</i> (top 3%) | Mar. 2023 |
| • Sailvan Times Scholarship (2%), <i>Harbin Institute of Technology</i> | Mar. 2023 |
| • National Scholarship (for Graduate Student, 2%) | Oct. 2021 |
| • NLPCC 2021 Shared Task: Argument Pair Extraction (1st Place) | Mar. 2021 |
| • SemEval 2021 Task 5: Toxic Spans Detection (1st Place, Team Lead) | Feb. 2021 |
| • Outstanding Bachelor's Graduate & Outstanding Bachelor's Thesis, <i>HIT</i> (top 3%) | Jun. 2020 |
| • National Scholarship (for Undergraduate Student, 2%) | Oct. 2019 |

SKILLS

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|---------------------------------|--|
| • LLM Training & Post-training: | Pre-training, continued pre-training, SFT, RLHF, GRPO, LoRA/PEFT |
| • Distributed Training: | DeepSpeed, FSDP, Megatron-LM |
| • Inference & Serving: | vLLM, SGLang |
| • Programming: | Python, C/C++, PyTorch |

ACADEMIC SERVICE

Reviewer for: ICML, NeurIPS, ICLR, ACL, AAAI, KDD, EMNLP, NAACL, NLPCC

TEACHING EXPERIENCE

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| • Teaching Assistant for Data Structure | King's College London, 2025 Spring |
| • Teaching Assistant for Data Science | King's College London, 2025 Spring |
| • Teaching Assistant for Natural Language Processing | Harbin Institute of Technology, 2020 Fall |